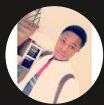




Infrastructure as Code with CloudFormation



Ahmed Tetteh

nextnetwork-devops-stack

Stack info

Events

Resources

Outputs

Parameters

Template

Change sets

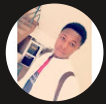
Git sync

Resources (16)

Q

Search resources

Logical ID	Physical ID	Type	Status
CodeArtifactDomainDomainnetwork	arn:aws:codeartifact:ap-northeast-1:007399901516:domain/network	AWS::CodeArtifact::Domain	CREATE_COMPLETE
CodeArtifactRepositoryRepositorynetworkmaincentralstore	arn:aws:codeartifact:ap-northeast-1:007399901516:repository/network/network-central-store	AWS::CodeArtifact::Repository	CREATE_COMPLETE
CodeArtifactRepositoryRepositorynetworknetworkdevopsclcd	arn:aws:codeartifact:ap-northeast-1:007399901516:repository/network/network-devops-clcd	AWS::CodeArtifact::Repository	CREATE_COMPLETE
CodeBuild00nextnetworkwebcloudformation00	network-devops-clcd	AWS::CodeBuild::Project	CREATE_COMPLETE
CodeDeployApplicationnextnetworkdevopsclcd	network-devops-clcd	AWS::CodeDeploy::Application	CREATE_COMPLETE
CodeDeployDeploymentGroup00nextnetworkwebcloudformation00	network-devops-clcd-deploymentgroup	AWS::CodeDeploy::DeploymentGroup	CREATE_COMPLETE
CodeStarConnectionsConnection03de2165b594741e2568bcb596536c	arn:aws:codestar-connections:ap-northeast-1:007399901516:connection/05b4129c-2466-44cc-b94a-07c6d842f9b1	AWS::CodeStarConnections::Connection	CREATE_COMPLETE
IAMInstanceProfileEC2instancenetworkclcd	ec2-instance-network-clcd	AWS::IAM::InstanceProfile	CREATE_COMPLETE
IAMManagedPolicyPolicycodeartifactnetworkconsumerpolicy	arn:aws:iam::007399901516:policy/codeartifact-network-consumer-policy	AWS::IAM::ManagedPolicy	CREATE_COMPLETE
IAMManagedPolicyPolicycodeartifactnetworkconsumerpolicy	arn:aws:iam::007399901516:policy/service-role/CodeBuildAWSPolicy-network-devops-clcd-ap-northeast-1	AWS::IAM::ManagedPolicy	CREATE_COMPLETE
IAMManagedPolicyPolicycodeartifactnetworkconsumerpolicy	arn:aws:iam::007399901516:policy/service-role/CodeBuildCloudWatchLogPolicy-network-devops-clcd-ap-northeast-1	AWS::IAM::ManagedPolicy	CREATE_COMPLETE
IAMManagedPolicyPolicycodeartifactnetworkconsumerpolicy	arn:aws:iam::007399901516:policy/service-role/CodeBuildCodeConnectionsSourceCredentialsPolicy-network-devops-clcd-ap-northeast-1	AWS::IAM::ManagedPolicy	CREATE_COMPLETE
IAMRoleCodeBuildnextnetworkdevopsclcdservice	codebuild-network-devops-clcd-service-role	AWS::IAM::Role	CREATE_COMPLETE
IAMRoleEC2instancenetworkclcd	ec2-instance-network-clcd	AWS::IAM::Role	CREATE_COMPLETE
IAMRoleNextWorkCodeDeployRole	NextWorkCodeDeployRole	AWS::IAM::Role	CREATE_COMPLETE
S3BucketNextWorkdevopsclcd	network-devops-clcd-fap	AWS::S3::Bucket	CREATE_COMPLETE



Introducing Today's Project!

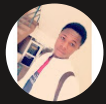
In this project, I will use Infrastructure as Code (IaC) generator to define and gather all my resources into a CloudFormation template and deploy them faster and easily, bringing about consistency and flexible management. I'm doing this project to learn the power of CloudFormation for delivering faster results.

Key tools and concepts

Services I used were CloudFormation, CodeDeploy and many more services. Key concepts I learnt include Infrastructure as Code, how to use the IaC generator and ways of troubleshooting errors when launching a CloudFormation template.

Project reflection

This project took me approximately 4 hrs, including resource identification in the IaC generator, error debugging during launch and how to manually define resources such as CodeBuild and CodeDeploy deployment group in the template. The most challenging part was editing the CloudFormation template and debugging the errors as they occurred. It was most rewarding to see the launched template and all the created resources via the hyperlinks. through some few clicks.

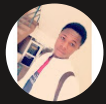


Ahmed Tetteh

NextWork Student

nextwork.org

This project is part six of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project CodePipeline in the next few hours.

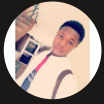


Generating a CloudFormation Template

The IaC Generator is tool in CloudFormation that simply takes inventory of all the existing resources (like EC2 and S3 buckets) within your AWS account that could be added to the CloudFormation template, enabling you to create your template faster. It works in a three-step process: 1. It first scans all the resources within your AWS account. 2. Creates a template using the resources it discovered to deploy and manage them together. 3. Finally, this template can then be imported into CloudFormation to deploy all the resources in one go.

A CloudFormation template is simple a text file that consists of all the bundled up resources you want to deploy and manage in code format. The resources that I could add to my template includes S3 bucket, CodeArtifact (domains & repositories), CodeDeploy(Applications), IAM (roles & policies) and CodeStar connections.

The resources I couldn't add to my template were Codebuild and CodeDeploy deployment group, due to their complex nature as they require very specific or flexible configurations. Hence, the IaC generator is unable to add them to the scanned resources. For the network resources, am choosing to opt out of them because, they will increase the complexity of my template.



🔔 You have successfully created the template. Next, you can migrate to CDK or import to a stack.

nextwork-devops-cicd

Template details

Template generation status

🟢 Complete

Template ID

🔗 [arn:aws:cloudformation:ap-northeast-1:007399901516:generatedTemplate/618ed4ce-1dad-49d7-9d54-2907f6f752f2](#)

▼ Configurations

Deletion policy

DELETE

Update replace policy

DELETE

Creation time

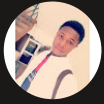
2025-12-30 17:41:36 UTC+0900

Updated time

2025-12-30 17:41:46 UTC+0900

Built off existing stack

-

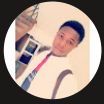


Template Testing

Before testing my template, I deleted the existing ones I created manually to prevent failure because, CloudFormation will create those new resources using the same names as the existing ones.

I tested my template by uploading it to CloudFormation to create a stack. The result of my first test was failed because, some of my policies and roles were being created at simultaneously, which should not be the case, as we know that policies are attached to the roles, then the roles are to be assumed by an entity or resource. Hence, the roles need to be generated first. Think of it like a folder-file relationship.

e8328a52-da53-4cd7-a431-31fe1954444e	2025-12-30 21:46:42 UTC+0900	CodeStarConnectionsConnectionConnection03de2165f6594741a2568bcd596536c	 CREATE_FAILED Likely root cause	Resource handler returned message: "Account 007399901516 does not have access to host arn:aws:codestar-connections:ap-northeast-1:000000000000:host/00000000-0000-0000-000000000000 (Service: CodeStarConnections, Status Code: 400, Request ID: 202f2865-24f0-48ea-8b5c-ae5dee4f7aea) (SDK Attempt Count: 1)" (RequestToken: e6c021f1-61a7-12a4-66b4-b35dcac939ff, HandlerErrorCode: GeneralServiceException)
--	------------------------------	--	--	--

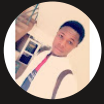


DependsOn

To resolve the error, I opened up the template in a code editor (VS Code) and added an extra line of code to my policies, i.e, "DependsOn". The DependsOn attribute means that am telling CloudFormation to wait on creating that resource because, that resource relies or depends on another resource to be available first.

The DependsOn line was added to four different parts of my template: the CodeBuild policies and the CodeArtifact policy... For CodeArtifact policy, since the policy is attached to two roles, i.e. the CodeBuild service role and the EC2 instance role, then the IDs of the two roles should be included in the DependsOn line.

```
    Name: "IAMManagedPolicyPolicyServiceRoleCodeBuildCodeConnectionsSourceCredentialsPolicynextworkdevopscidapnortheast1007399901516:"  
    UpdateReplacePolicy: "Delete"  
    Type: "AWS::IAM::ManagedPolicy"  
    DeletionPolicy: "Delete"  
    DependsOn: "IAMRoleCodebuildnextworkdevopscidserviceRole"  
    Properties:
```



Circular Dependencies

I gave my CloudFormation template another test! But this time I got a different error message, i.e. "Circular dependency error between resources" error. This error signifies a chicken a egg situation, whereby from the template, CloudFormation senses that some resources are stuck in a loop. In my case, this is between my IAM roles and their policies, as both are seen to be waiting on eachother to be created first, causing this circular dependency error.

To fix this error, I deleted the references within the IAM roles pointing to the policies in the ManagedPolicyARN section. So, now CloudFormation should just create the role first, then the policies, and attach them to the specific role during creation.

Specify template info
This [GitHub repository](#) contains sample CloudFormation templates that can help you get started on new infrastructure projects. [Learn more](#)

Template source
Selecting a template generates an Amazon S3 URL where it will be stored. A template is a JSON or YAML file that describes your stack's resources and properties.

☐ Amazon S3 URL
Provide an Amazon S3 URL to your template.

☒ Upload a template file
Upload your template directly to the console.

☐ Sync from Git
Sync a template from your Git repository.

Upload a template file

Choose file

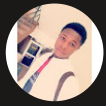
Circular dependency between resources: [IAMManagedPolicyPolicyServiceRoleCodeBuildCodeConnectionsSourceCredentialsPolicynextworkdevopscidapnortheast1007399901516, IAMManagedPolicyPolicyServiceRoleCodeBuildCloudWatchLogsPolicynextworkdevopscidapnortheast1, IAMInstanceProfileEC2Instancenextworkcid, IAMManagedPolicyPolicyServiceRoleCodeBuildBasePolicynextworkdevopscidapnortheast1, IAMManagedPolicyPolicycodeartifactnextworkconsumerpolicy, IAMRoleEC2Instancenextworkcid, IAMRoleCodeBuildnextworkdevopscidserviceRole]

JSON or YAML formatted file

S3 URL: <https://s3.ap-northeast-1.amazonaws.com/cf-templates-uwvjryzej7nc-ap-northeast-1/2025-12-30T204147.891Z18c-nextwork-devops-cicd-template-1767094259816.yaml>

View in Infrastructure Composer

Circular dependency between resources: [IAMManagedPolicyPolicyServiceRoleCodeBuildCodeConnectionsSourceCredentialsPolicynextworkdevopscidapnortheast1007399901516, IAMManagedPolicyPolicyServiceRoleCodeBuildCloudWatchLogsPolicynextworkdevopscidapnortheast1, IAMInstanceProfileEC2Instancenextworkcid, IAMManagedPolicyPolicyServiceRoleCodeBuildBasePolicynextworkdevopscidapnortheast1, IAMManagedPolicyPolicycodeartifactnextworkconsumerpolicy, IAMRoleEC2Instancenextworkcid, IAMRoleCodeBuildnextworkdevopscidserviceRole]



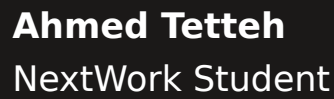
Manual Additions

In a project extension, I manually defined two more resources: the CodeBuild project, and the CodeDeploy deployment group.

I also had to make sure the references were consistent in this template, so I put in the IDs for the service roles, s3 bucket and CodeDeploy Application from the template, making sure they were consistent across all fields.

I also introduced Parameters, which are reusable, flexible parts of the template that allows you to parametrize values for different use cases without actually modifying the template itself. This is very useful for a cases where we have to reference different GitHub account info that can be easily changed for a different working environment. Also, with parameters, we can request CloudFormation to wait for a user input before creating the resources.

```
# CodeBuild Project
CodeBuild00nextworkwebcloudformation00:
  Type: AWS::CodeBuild::Project
  DependsOn:
    - "IAMRoleCodebuildnextworkdevopsicdserverole"
    - "S3BucketNextworkdevopsicdfap"
  Properties:
    Name: nextwork-devops-cicd
    Description: Build project for NextWork web application
    Source:
      Type: GITHUB
      Location: !Sub "https://github.com/${GitHubRepoOwner}/${GitHubRepo}" Unresolved tag: !Sub
      BuildSpec: buildspec.yml
    Artifacts:
      Type: S3
      Name: nextwork-web-build.zip
      Packaging: ZIP
      Location: !Ref S3BucketNextworkdevopsicdfap Unresolved tag: !Ref
      Path: /builds
    Environment:
      Type: LINUX_CONTAINER
      ComputeType: BUILD_GENERAL1_SMALL
      Image: aws/codebuild/amazonlinux2-x86_64-standard:corretto8
      ServiceRoles: !GetAtt IAMRoleCodebuildnextworkdevopsicdserverole.Arn Unresolved tag: !GetAtt
    LogsConfig:
      CloudWatchLogs:
        GroupName: /aws/codebuild/nextwork-devops-cicd
        Status: ENABLED
        StreamName: webapp
```



Success!

I could verify all the deployed resources by visiting hyperlinks for all the created resources in the Resources tab of my CloudFormation Stack panel.

[illegible]



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